Research Proposal

Chemical Bonding Limits and Electron–Phonon Optimization in Hydrogen Clathrate Superconductors

Background

Hydrogen-rich clathrate superconductors represent the most successful class of phonon-mediated superconductors discovered to date. Binary, ternary, and quaternary hydrides have produced superconducting transition temperatures (Tc) approaching room temperature under high pressure. Nevertheless, despite extensive crystal-structure prediction, high-throughput screening, and experimental synthesis, essentially all known high-Tc hydrides remain confined to approximately 200–300 K. The persistence of this apparent ceiling suggests that Tc is controlled by a fundamental physical and chemical limit rather than simply by composition.

Previous work from my group demonstrated, through the functional derivative of the critical temperature with respect to the Eliashberg spectral function, $\delta Tc/\delta\alpha^2F(\omega)$, that the phonons contributing most efficiently to superconductivity are low-frequency anharmonic hydrogen translational modes together with intermediate-frequency H–H–H bending modes (K. Tanaka, J. S. Tse, and H. Liu, Electron-phonon coupling mechanisms for hydrogen-rich metals at high pressure, Phys. Rev. Rapid Comm. 96, 100502(R) 2017). Further analysis based on the Midgal Eliashberg extended BCS theory (J.P. Carbotte, Properties of boson-exchange superconductors. Rev Mod Phys,1990; 62: 1027–157) I revealed revealed an approximately linear relationship between the superconducting gap, Tc, and the characteristic phonon frequency, with an optimum average frequency close to 1100 cm^{-1} (J.S. Tse, A Chemical Perspective on High Pressure Crystal Structures and Properties, Nat. Sci. Rev., 7, 149, 2020). These findings imply that superconductivity is fundamentally governed by the properties of the hydrogen framework rather than the identity of the metal atom.

The central premise of this proposal is that the hydrogen clathrate network can only be weakened to a finite extent before structural instability occurs. Electron transfer from the metal simultaneously enhances electron–phonon coupling and weakens H–H bonding. The highest attainable Tc is therefore expected to emerge from a balance between lattice stability and electron–phonon optimization. Establishing this balance is the central scientific objective of the proposed research.

Research Hypothesis

There exists a universal chemical-bonding limit that determines the maximum superconducting transition temperature attainable in hydrogen-clathrate superconductors. This limit is governed by the interplay between charge transfer, H–H bond strength, lattice dynamics, and electron–phonon coupling, and is largely independent of chemical composition.

Research Objectives

1. Establish quantitative descriptors for H–H bonding that define the stability of hydrogen clathrate frameworks.
2. Determine how charge transfer modifies hydrogen bonding and vibrational spectra.
3. Identify the optimum phonon frequencies that maximize Tc using Eliashberg functional-derivative analysis.
4. Develop a universal relationship linking chemical bonding, phonon frequencies, superconducting gap, and Tc.

Research Packages

Quantitative chemical bonding.

Orbital-projection methods such as COHP/LOBSTER and conventional population analyses become increasingly unreliable under extreme compression because the atomic basis functions lose transferability and projection errors become significant (see the discussion in the peer review file of this paper, https://www.nature.com/articles/s42004-024-01245-9). Instead, this project will employ electron-density-based descriptors. Quantum Theory of Atoms in Molecules (QTAIM) will be used to determine bond critical points, electron density, Laplacian, energy density, and bond-basin populations. Complementary Density Derived Electrostatic and Chemical (DDEC6) bond orders will provide chemically meaningful bond-order estimates. Together these quantities will establish quantitative criteria defining stable hydrogen frameworks.
In addition, the electronic band structure of the hypothetical H-clathrates will be examined to determine the orbital characteristics of the extra electrons donated by the encaged metal. I suggested in the CaH_6 study that the electrons from Ca are added to a degenerate band, resulting in electronic instability.

Stability of the hydrogen clathrate network.

Representative hydrogen clathrate frameworks will be stripped of the metal atoms to produce hypothetical empty hydrogen cages. The electronic charge on the framework will then be varied systematically while maintaining overall neutrality using a compensating positive jellium background. Phonon calculations performed at each charge state will follow the evolution of translational, librational, stretching, and H–H–H bending modes and identify the onset of dynamical instability. These calculations directly determine the maximum charge that can be accommodated by the hydrogen framework before collapse and therefore provide chemically meaningful design rules for selecting electron-donating metal atoms.

Electron–phonon optimization.

I have developed codes to solve the isotropic Eliashberg equations based on iterative and matrix-diagonalization solvers will compute Tc, the zero-temperature superconducting gap, and the functional derivative $\delta T_c/\delta\alpha^2F(\omega)$. The functional derivative identifies the frequency regions where additional electron–phonon coupling produces the greatest enhancement of Tc. By repeating this analysis across representative clathrate families, the project will determine whether a universal optimum phonon frequency exists and whether the previously identified value near 1100 cm^{-1} represents a fundamental limit.

Validation and predictive framework.

The simplified Eliashberg methodology will be benchmarked against EPW calculations within both isotropic and anisotropic formulations for representative systems. The combined data set, including bond descriptors, charge transfer, phonon spectra, optimized frequencies, superconducting gaps, and Tc, will be used to derive predictive relationships connecting crystal chemistry directly with superconducting performance.

Expected Outcomes

The project will (i) establish quantitative chemical criteria for hydrogen-framework stability, (ii) determine the maximum charge transfer sustainable by hydrogen clathrates, (iii) identify the universal optimum phonon frequency for phonon-mediated superconductivity, (iv) explain the observed saturation of Tc among known hydrides, and (v) provide predictive design principles for discovering next-generation hydrogen-rich superconductors.

Significance

The novelty of this work lies in shifting the emphasis from searching for new hydrides toward understanding the fundamental physical limit of superconductivity in hydrogen clathrates. By integrating chemical bonding theory, lattice dynamics, and Eliashberg superconductivity within a single quantitative framework, the project will establish a mechanistic theory that explains existing observations while providing predictive guidance for future materials discovery.